Pinned group summary: `SO(n=4, q=2)`

Pinning-Sympy automated output

July 6, 2026

Contents

1 Basic group attributes

Group	`SO(n=4, q=2)`
Matrix size	4×4
Rank	2
Defining equations	$DefiningEquationPlaceholder$

1.1 Symmetric bilinear form

Dimension	4
Witt index	2
Primitive element	N/A
Epsilon	N/A

Matrix: $\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$

1.2 Generic Lie algebra element

$$\begin{bmatrix} x_{00} & x_{01} & 0 & -x_{12} \\ x_{10} & x_{11} & x_{12} & 0 \\ 0 & -x_{30} & -x_{00} & -x_{10} \\ x_{30} & 0 & -x_{01} & -x_{11} \end{bmatrix}$$

1.3 Generic torus element

$$\begin{bmatrix} t_0 & 0 & 0 & 0 \\ 0 & t_1 & 0 & 0 \\ 0 & 0 & \frac{1}{t_0} & 0 \\ 0 & 0 & 0 & \frac{1}{t_1} \end{bmatrix}$$

1.4 Trivial characters

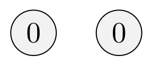
$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

2 Root system

Summary Properties

Dynkin type	`A1_x_A1`
Irreducible	False
Reduced	True
Simply laced	True
Number of roots	4

2.1 Dynkin diagram



2.2 Irreducible components

This is a reducible root system with the following components:

Dynkin type	**Rank**
`A1`	1
`A1`	1

2.3 Roots and coroots

Root	**Coroot**	**Simple**	**Sign**	**Multipliable**
$(-1,\ -1,\ 0,\ 0)$	$(-1,\ -1,\ 1,\ 1)$	No	$-$	No
$(-1,\ 1,\ 0,\ 0)$	$(-1,\ 1,\ 1,\ -1)$	No	$-$	No
$(1,\ -1,\ 0,\ 0)$	$(1,\ -1,\ -1,\ 1)$	Yes	$+$	No
$(1,\ 1,\ 0,\ 0)$	$(1,\ 1,\ -1,\ -1)$	Yes	$+$	No

2.4 Linear combinations of roots

Definition 2.1. For two roots α, β in a root system Φ, the set of positive integral linear combinations of α, β that are also roots is denoted $(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$

No pairs of roots generate positive integer linear combinations.

3 Root spaces

Root	Dimension
$(-1,\ -1,\ 0,\ 0)$	1
$(-1,\ 1,\ 0,\ 0)$	1
$(1,\ -1,\ 0,\ 0)$	1
$(1,\ 1,\ 0,\ 0)$	1

RootSpaceTablePlaceholder

4 Root subgroups

RootSubgroupTablePlaceholder
HomomorphismDefectTablePlaceholder
HomomorphismDefectIdentitiesPlaceholder

5 Commutators

CommutatorCoefficientTablePlaceholder
CommutatorIdentitiesPlaceholder

6 Weyl group

WeylGroupPlaceholder
WeylConjugationCoefficientPlaceholder
WeylConjugationIdentitiesPlaceholder

7 Coroot torus elements (h_α)

CorootTorusElementPlaceholder